MARKETGUARD AI: MULTI-AGENT STOCK SPECULATION & MANIPULATION DETECTION SYSTEM

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Abstract. Stock manipulation and speculation pose substantial challenges for investors, financial firms, and regulatory bodies since there are instances where there may be suspicious actions that are complicated and hard to tell apart from normal market fluctuations. The paper introduces the MarketGuard AI: Multi-Agent Stock Speculation & Manipulation Detection System, a machine learning approach to detect any suspicious market actions through financial time series and multiple stock interactions. The current work centers around the use of Spatio-Temporal Graph Transformer (ST-GT) as the main algorithm for detecting suspicious activities in the market. Data from stocks A, AAPL, and AE is analyzed to derive financial metrics such as return, volume delta, moving average delta, RSI, volatility, and volume anomalies. Proxy suspicious-activity tagging relies on extreme standardized returns and abnormal trading volume. Each stock is treated as a node of dynamically generated graphs whose relationships are based on the rolling return correlation. A Graph Attention Network learns the interactions between stocks, while Transformer components model temporal dynamics and market behavior. The model is trained with a train-validation-test time series split, applying weighted binary cross-entropy loss to tackle class imbalance. In the experimental analysis, ST-GT achieved the accuracy of 90.94%, the precision of 6.51%, the recall of 18.97%, the F1 score of 9.69%, and the ROC-AUC of 65.38%. The results show that graph-based spatio-temporal analysis can find important market behaviors, but low precision and recall imply the necessity of further improvements. The next phase of the work includes extending the developed prototype with a Temporal Fusion Transformer for risk prediction and a Multi-Agent Large Language Model based evidence fusion and explanations.

Keywords: Stock market surveillance; market manipulation; anomaly detection; graph neural networks; graph attention; Transformer; spatio-temporal learning; financial time series; ST-GT

1 Introduction

Financial markets create large numbers of time-dependant and inter-connected observations and thus make it challenging to detect any suspicious activities within the

financial markets using individual time series model alone. Market manipulation may be a result of strange combination of price action, trading volumes, volatility and relation between correlated securities. Current studies on the topic therefore tend to utilise the deep learning, graph learning, attention mechanism and temporal methods to capture dependencies between securities and changes in market behaviour. MarketGuard AI approaches this problem by creating a dynamic stock graph based on rolling correlation and combining Graph Attention with Transformer encoder. The aim of this system is not to declare the transaction to be fraudulent but to provide suspicious activity warning which can be used for further investigation. Provided datasets include historical OHLCV information for three stocks and there are no manipulation labels provided in the dataset; hence, the currently developed prototype will have statistically generated anomaly label.

2 Literature Survey

The survey below uses 15 research works selected for their relevance to market-manipulation detection, stock selection, dynamic stock relationships, graph learning, and temporal attention. Five are treated as core papers for the present project direction: Khodabandehlou and Golpayegani; Liu, Li, and Shi; Safa et al.; Liao et al.; and Dai et al. The remaining works provide the spatio-temporal graph and Transformer background needed to position the ST-GT design.

1. Khodabandehlou and Golpayegani (2022): Their systematic review maps trade-based market manipulation detection methods, anomaly-based formulations, data-mining approaches, and open challenges. The major gap relevant to MarketGuard AI is the scarcity of reliable labelled manipulation data and the difficulty of representing evolving market relationships. This motivates a risk/anomaly formulation and careful separation between suspiciousness and legally verified manipulation.
2. Wang et al. (2022): The Adaptive Long-Short Pattern Transformer (ALSP-TF) addresses non-stationary stock selection by learning fine-grained short- and long-term patterns and using graph self-supervised regularization. Its importance to MarketGuard is the use of adaptive attention and stock-level relational learning instead of treating securities independently.
3. Li et al. (2022): Chart GCN converts technical-chart information into graph structures and applies graph kernels/GCN for stock movement prediction. It demonstrates that graph representations can capture information that is difficult to express with conventional scalar technical indicators, supporting the use of graph learning in financial time series.
4. Zulkifley et al. (2023): This survey reviews AI-based stock-market manipulation detectors across conventional machine learning, deep learning, CNN/RNN methods, and manipulation taxonomies. It reinforces the need for scalable automated surveillance and identifies limitations caused by complex manipulation patterns and large trading volumes.
5. Huynh et al. (2023): The WSDM work models both multi-order stock dynamics and individual stock behaviour. Its emphasis on non-pairwise relationships is directly

relevant to graph-based MarketGuard modelling, because suspicious activity may involve coordinated movements among multiple securities rather than one isolated series.

6. Xia et al. (2024): CI-STHPAN combines Transformer and hypergraph-attention representations with self-supervised pretraining and dynamic stock relationships. The work demonstrates that predefined static relations can be inadequate in volatile markets. MarketGuard adopts the same broad principle through dynamically recomputed correlation graphs, while using a lighter GAT-plus-Transformer implementation.

7. Liao et al. (2024): DHSTN introduces dynamic hypergraph construction, industry relation aggregation, and temporal modelling for stock trend prediction. The key contribution for MarketGuard is the explicit treatment of changing stock relationships. Our prototype simplifies the relation structure to a dynamic pairwise graph generated from rolling return correlations.

8. Li et al. (2024): MASTER, or Market-Guided Stock Transformer, models momentary and cross-time stock correlations and uses market information for feature selection. It demonstrates the value of Transformer architectures that explicitly represent cross-stock dependencies rather than applying a temporal model to each stock independently.

9. Yan and Tan (2024): DPA-STIFormer proposes adaptive-correlation spatial-temporal inverted Transformer modelling for stock time series. It separates temporal and feature representations and fuses multiple correlation views. This work supports the project hypothesis that stock relationships should be learned dynamically rather than fixed permanently.

10. Gu et al. (2024): FinBERT-LSTM integrates financial-news sentiment with historical stock information. The study shows the value of combining textual market information with numerical time series. For MarketGuard, this suggests a future extension in which a news agent supplies external evidence to complement the ST-GT market-data agent.

11. Liu, Li, and Shi (2024): This paper proposes a stacking-based neural ensemble for stock price manipulation detection using data derived from China Securities Regulatory Commission announcements. It is especially relevant because it uses manipulation-specific regulatory data rather than only forecasting labels. The paper motivates MarketGuard's future transition from proxy anomaly labels to verified manipulation-event labels.

12. Safa et al. (2024): WALDATA applies wavelet transformation and adversarial learning to anomalous trading-activity detection. The study highlights the difficulty of detecting rare, dynamically evolving manipulation signals and the scarcity of labelled anomalies. This supports MarketGuard's current anomaly-oriented formulation and points toward frequency-domain features as a future improvement.

13. Liu et al. (2025): The Market-wide Dynamic Hypergraph Attention Network (MDHAN) combines market-wide temporal influence with dynamic hypergraph spatial dependencies. It extends stock modelling beyond stock-specific information and is relevant to MarketGuard's goal of detecting coordinated market behaviour.

14. Wu et al. (2025): STHformer uses a spatio-temporal hypergraph Transformer with self-supervised pretraining for multi-stock trend forecasting. It directly supports the

use of Transformer-based temporal modelling together with dynamic higher-order stock relationships and shows the research trend toward pretrained spatio-temporal financial models.
15. Dai et al. (2026): SD-FMM addresses financial market manipulation through self-supervised anomaly synthesis, domain-specific features, and a dual-branch contrastive detection network. It is highly relevant to the next stage of MarketGuard because it explicitly targets manipulation detection under label scarcity and boundary ambiguity. Its direction suggests a strong future novelty: combine dynamic graph-temporal evidence with self-supervised manipulation-event synthesis and multi-agent evidence fusion.

2.1 Literature Gap and Project Positioning

Limitations that have been repeated throughout the literature include (1) that many stock models prioritize prediction or ranking over surveillance; (2) that manipulation-specific data sets are few and unbalanced; and (3) that relationships between stocks are either pre-defined or considered constant. The novelty of the MarketGuard project is that it uses a graph constructed in real-time and temporal Transformers for the purpose of calculating suspicious activity scores, using the current label as an anomaly as part of its public data set. The future research contribution will be to use verified events of manipulation in place of this proxy, and to enrich the ST-GT system by adding news/evidence agents and evidence fusion to the core model.

2.2 Five Core Papers Used for the Project Direction

Table 1. Five core papers used for the project direction.

Paper	Focus	Method	Novelty / relevance
Khodabandehlou & Golpayegani, 2022	Systematic review of market manipulation detection	Manipulation taxonomy, anomaly framing, data-mining approaches	Identifies label scarcity and open surveillance challenges.
Liu, Li & Shi, 2024	Stock-price manipulation detection with stacking ensemble	RNN-based submodules + neural stacking on CSRC-derived data	Uses manipulation-specific regulatory evidence and preserves learned neural features.
Safa et al., 2024	WALDATA	Wavelet transformation + adversarial learning	Targets rare and dynamically evolving anomalous trading activity.
Liao et al., 2024	DHSTN	GRU + dynamic hypergraph + industry relation aggregation	Learns changing multi-stock relationships instead of relying only on fixed pairwise links.
Dai et al., 2026	SD-FMM	Self-supervised anomaly synthesis + domain features + dual-branch contrastive network	Addresses label scarcity, concealed signals, and manipulation-boundary ambiguity.

3 Proposed Methodology

The corrected prototype receives daily OHLCV records for three stocks, engineers seven financial features, aligns the securities on common trading dates, creates 20-day

temporal windows, and builds a dynamic graph from rolling return correlations. The seven features are daily return, volume change, SMA-5 gap, SMA-20 gap, RSI-14, 20-day volatility, and 20-day volume z-score. Features are projected to 32 dimensions, processed spatially with a Graph Attention layer, and then processed stock-wise through a two-layer, four-head Transformer. A stock-wise sigmoid probability is converted to a 0–100 risk score for presentation.

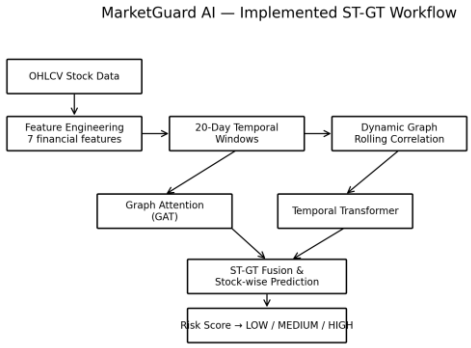


Fig. 1. Implemented MarketGuard AI ST-GT workflow.

Fig. 1 shows the implemented data path. Raw OHLCV records are converted into technical and statistical features, then grouped into 20-day windows. Rolling return correlations determine dynamic graph edges. The GAT learns spatial inter-stock dependencies, while the Transformer learns temporal dependencies. The two stages are sequentially combined into the ST-GT representation, which produces stock-wise suspicious-activity probabilities and the final risk categories.

4 Algorithm

Input: OHLCV time series D for N stocks; window length W=20.

Clean dates, sort chronologically, remove duplicate dates and invalid rows, and retain common dates across stocks.

For each stock, compute return, volume change, SMA-5 gap, SMA-20 gap, RSI-14, 20-day volatility, and 20-day volume z-score.

Construct a proxy target y_t for suspicious activity when the next-day absolute return z-score exceeds 3 or the next-day volume z-score exceeds 3.

Split the observations chronologically into 70% training, 15% validation, and 15% test data; fit the StandardScaler only on the training portion.

For every 20-day window, calculate the return correlation matrix and connect each stock to its strongest absolute-correlation neighbour; symmetrize the graph.

Project the seven input features to a 32-dimensional latent space and apply Graph Attention to aggregate information from dynamically connected neighbouring stocks.

Rearrange the graph-enhanced representations into stock-wise temporal sequences and process each sequence with a two-layer, four-head Transformer encoder.

Use the final temporal representation for each stock and apply LayerNorm followed by a linear classifier.

Train with weighted BCEWithLogitsLoss and AdamW for five epochs in the classroom prototype, using gradient clipping; select a classification threshold on the validation set and evaluate the held-out test set using accuracy, precision, recall, F1-score, and ROC-AUC.

Convert the latest predicted probability p into a risk score 100p and map it to LOW, MEDIUM, or HIGH.

Output: stock-wise suspicious-activity risk table and evaluation diagnostics.

5 Dataset Description and Preprocessing

The uploaded datasets contain daily OHLCV records for A (5,124 rows; 1999-11-18 to 2020-04-01), AAPL (10,171 rows; 1984-09-07 to 2025-01-17), and AE (11,387 rows; 1975-02-07 to 2020-04-01). The notebook uses A, AAPL, and AE; aapl_us_d.csv is not used. Dates are parsed and sorted, duplicates removed, and common dates aligned. Seven engineered features are standardized using training data only. The resulting common dataset contains 5,025 usable dates and a 2.24% proxy suspicious rate.

6 Mathematical Formulation of the Proposed Model

The model begins with financial feature construction. For stock i and trading day t, the core price and volume features are:

$$r_{i,t} = \frac{C_{i,t}}{C_{i,t-1}} - 1$$

Fig. 5. Daily return formula.

$$\Delta V_{i,t} = \frac{V_{i,t}}{V_{i,t-1}} - 1$$

Fig. 6. Volume change formula.

$$G_{i,t}^{(k)} = \frac{C_{i,t}}{SMA_{i,t}^{(k)}} - 1, \qquad SMA_{i,t}^{(k)} = \frac{1}{k}\sum_{j=0}^{k-1} C_{i,t-j}$$

Fig. 7. Moving-average gap formula.

$$\sigma_{i,t}^{(20)} = \mathrm{std}(r_{i,t-19}, \ldots, r_{i,t})$$

Fig. 8. Rolling volatility formula.

$$z^{V}_{i,t} = \frac{V_{i,t} - \mu^{V}_{i,t,20}}{\sigma^{V}_{i,t,20} + \varepsilon}$$

Fig. 9. Volume z-score formula.

$$z^{r}_{i,t} = \frac{r_{i,t} - \mu^{r}_{i,t,20}}{\sigma^{r}_{i,t,20} + \varepsilon}$$

Fig. 10. Return z-score formula.

$$y_{i,t} = \mathbf{1}(|z^{r}_{i,t+1}| > 3 \ \vee \ z^{V}_{i,t+1} > 3)$$

Fig. 11. Proxy label formula.

$$A_t(i,j) = 1 \text{ if } j \text{ is among the strongest } k \text{ absolute-return correlations of } i; \quad A_t(i,i) = 1$$

Fig. 12. Dynamic graph formula.

$$h'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j, \qquad \alpha_{ij} = \mathrm{softmax}_j(\mathrm{LeakyReLU}(a_s^{\top} W h_i + a_d^{\top} W h_j))$$

Fig. 13. Graph attention formula.

$$H_{i,1:T} = \mathrm{Transformer}(h'_{i,1:T})$$

Fig. 14. Temporal Transformer formula.

$$\hat{y}_{i,t} = \sigma(w^{\top} LN(H_{i,T}) + b)$$

Fig. 15. Prediction formula.

$$\mathcal{L} = -w_+ y \log \hat{y} - (1-y)\log(1-\hat{y})$$

Fig. 16. Weighted binary loss formula.

$$R_i = 100\hat{y}_i$$

Fig. 17. Risk score formula.

The graph is dynamic because A_t is recomputed for each temporal window rather than fixed for the entire dataset. The GAT therefore learns a spatial representation conditioned on the current market relationship structure, while the two-layer, four-head Transformer models temporal dependencies for each stock.

7 Result Analysis

The original notebook did not contain saved execution outputs, so the corrected 27-cell implementation was reproduced from the uploaded A, AAPL, and AE datasets with a fixed random seed of 42. Over five epochs, training loss changed from 1.3458 to 1.2785 and validation loss changed from 1.2259 to 1.2226. A threshold of 0.65 was selected using validation F1. On the held-out test set, the corrected ST-GT achieved 90.94% accuracy, 6.51% precision, 18.97% recall, 9.69% F1-score, and 0.6538 ROC-AUC. The much lower recall than the earlier classroom run reflects the validation-selected threshold and the difficulty of the rare-event proxy task; precision also remains low, so the output should be interpreted as a screening signal, not a legal manipulation finding.

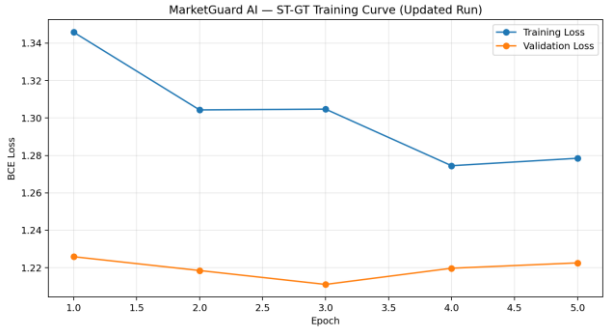


Fig. 2. Updated training and validation loss for the corrected five-epoch ST-GT run.

7.1 Result Table

Table 2. Experimental result comparison.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
ST-GT (corrected notebook)	0.9094	0.0651	0.1897	0.0969	0.6538
Logistic Regression (auxiliary baseline)	0.6485	0.0569	0.7931	0.1061	0.7832
Random Forest (auxiliary	0.9401	0.0978	0.1552	0.1200	0.7283

baseline)

Comparison note: Logistic Regression and Random Forest are retained as auxiliary reference baselines from the earlier paper version and are not part of the uploaded ST-GT notebook. Their values are shown for contextual comparison only; a fully controlled baseline study should be rerun under the exact final experimental protocol before publication.

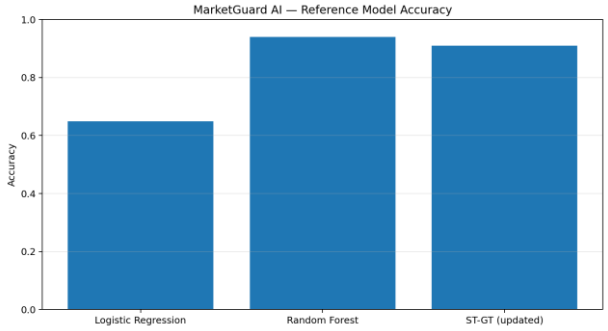


Fig. 3. Reference model accuracy comparison; auxiliary baselines are shown for context.

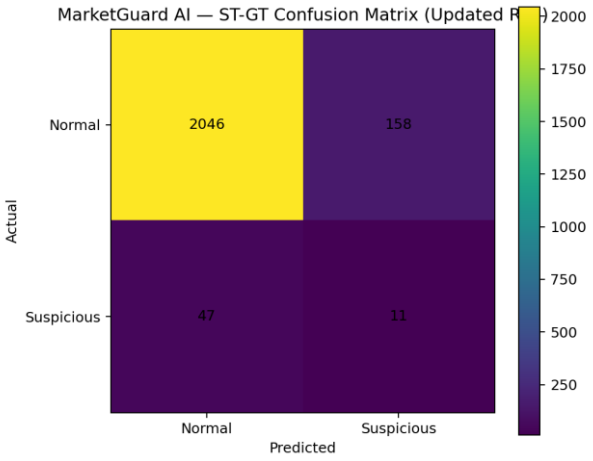


Fig. 4. Updated ST-GT confusion matrix for the corrected reproduced test run.

7.2 Latest MarketGuard AI Output

Table 3. Latest MarketGuard AI risk output.

Stock	Risk Score	Risk Level	Model
A	26.95	LOW	ST-GT
AAPL	26.16	LOW	ST-GT
AE	25.97	LOW	ST-GT

The corrected latest-window output assigns low suspicious-activity risk to all three stocks: A = 26.95, AAPL = 26.16, and AE = 25.97. These scores are model probabilities scaled to 100 and should be interpreted as anomaly-risk signals, not evidence of illegal manipulation. The current demonstration uses heuristic presentation bands of LOW < 40, MEDIUM 40–69.99, and HIGH ≥ 70.

8 Conclusion

MarketGuard AI demonstrates a workable ST-GT prototype that combines dynamic stock graphs and temporal attention for suspicious-activity scoring. Future work should add verified manipulation labels, news evidence, multi-agent fusion, and stronger baselines.

9 Limitations and Future Work

- Current datasets contain no verified manipulation labels; the target is a proxy anomaly rule.
- Only three stocks are used in the supplied implementation, limiting network-scale generalization.
- The current notebook implements the ST-GT core only; Multi-Agent LLM and TFT modules remain future extensions.
- The risk presentation thresholds (40 and 70) are heuristic and should not be confused with the learned classification threshold of 0.65; both require calibration using verified events.
- Future work should add regulatory event data, order-book or transaction-level data, financial-news sentiment, and explainable evidence fusion.
- Experiments should include multiple random seeds, longer training, more assets, ablation studies, and statistical significance tests.

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